

LAUPHEIM, Germany

WMO: 108370

Lat: 48.2175N

Lon: 9.9097E

Elev: 550

StdP: 94.89

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-8.9	-13.1	1.3	-10.9	-10.7	1.6	-7.9	13.2	7.5	11.5	7.0	2.2	60	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	30.1	19.2	28.1	18.5	26.2	17.8	20.1	27.7	19.2	26.4	18.4	25.0	2.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	13.3	22.7	16.8	12.8	22.1	16.0	12.1	21.1	59.8	28.0	56.9	26.7	54.2	25.0	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.1	6.8	DB	-15.1	32.9	3.8	1.7	-17.8	34.2	-20.1	35.2	-22.3	36.2	-25.1	37.5	(4)
(5)			WB	-15.1	21.6	3.8	0.9	-17.8	22.2	-20.0	22.7	-22.2	23.2	-24.9	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-0.3	0.9	4.7	8.9	13.2	16.8	18.6	18.3	13.8	9.5	4.0	0.9	(6)	
	DBStd	7.73	4.56	4.75	4.13	3.87	3.55	3.70	3.21	3.11	3.32	3.84	3.81	4.08	(7)	
	HDD10.0	1343	319	254	167	66	11	1	0	0	6	56	181	281	(8)	
	HDD18.3	3463	578	487	422	284	162	72	37	41	140	273	428	539	(9)	
	CDD10.0	1038	0	0	4	32	111	205	266	257	120	42	3	0	(10)	
	CDD18.3	117	0	0	0	0	4	27	44	38	4	0	0	0	(11)	
	CDH23.3	1251	0	0	0	3	67	296	446	389	48	2	0	0	(12)	
	CDH26.7	341	0	0	0	0	8	79	135	116	4	0	0	0	(13)	
(14) Wind	WSAvg	2.9	3.3	3.3	3.5	2.9	2.7	2.5	2.6	2.4	2.5	2.6	2.9	3.3	(14)	
(15) Precipitation	PrecAvg	888	49	43	51	63	101	116	107	99	74	63	62	59	(15)	
	PrecMax	1151	105	90	155	127	172	200	193	185	137	127	128	131	(16)	
	PrecMin	684	5	17	15	14	22	60	27	45	31	9	3	4	(17)	
	PrecStd	107	26	21	29	29	43	36	39	38	30	30	30	29	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	14.2	18.8	23.8	27.8	31.8	32.9	32.8	27.1	22.1	15.8	11.2	(19)	
		MCWB	8.2	8.3	10.6	13.5	17.1	19.5	19.9	19.4	17.8	15.4	11.9	8.2	(20)	
	2%	DB	9.6	11.3	16.0	21.1	25.2	29.0	30.0	29.6	24.8	19.2	13.0	9.4	(21)	
		MCWB	7.0	7.0	9.3	12.2	16.4	19.1	19.4	18.9	17.2	14.4	9.8	7.3	(22)	
	5%	DB	7.5	9.2	13.4	18.8	22.9	26.8	27.9	27.4	22.2	17.0	10.9	8.1	(23)	
		MCWB	5.5	6.0	8.4	11.5	15.1	18.2	18.7	18.2	16.1	13.3	8.8	6.3	(24)	
	10%	DB	5.8	7.2	11.2	16.7	20.8	24.7	25.8	25.2	20.1	15.0	9.3	6.5	(25)	
		MCWB	4.3	4.7	7.4	10.4	14.0	17.3	17.9	17.6	15.1	12.4	7.5	5.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.0	11.5	14.4	18.4	21.1	21.4	20.6	19.0	16.1	11.9	9.0	(27)	
		MCDB	10.9	13.5	16.8	21.6	25.4	29.0	29.5	28.6	25.0	20.4	14.6	10.7	(28)	
	2%	WB	7.1	7.4	10.0	13.0	17.0	19.8	20.1	19.6	17.7	14.8	10.3	7.6	(29)	
		MCDB	9.3	10.6	14.7	19.8	23.9	26.9	27.8	27.4	23.2	18.3	12.5	9.2	(30)	
	5%	WB	5.6	6.1	8.9	11.8	15.7	18.7	19.2	18.8	16.6	13.7	9.0	6.3	(31)	
		MCDB	7.1	9.0	12.8	17.6	21.7	25.6	26.5	25.8	21.6	16.6	10.8	7.8	(32)	
	10%	WB	4.2	4.8	7.7	10.7	14.5	17.6	18.3	18.0	15.5	12.6	7.7	5.0	(33)	
		MCDB	5.5	7.1	11.1	15.9	19.7	23.7	24.7	24.3	19.3	14.8	9.1	6.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	7.1	8.9	10.6	10.6	10.9	10.8	11.0	9.8	8.1	5.5	4.8	(35)	
		MCDBR	7.3	10.6	13.4	15.2	15.2	15.1	15.1	15.2	13.8	11.3	8.8	6.7	(36)	
	5% WB	MCWBR	5.4	7.0	8.0	8.0	7.5	6.9	6.2	6.4	7.3	6.9	6.2	5.1	(37)	
		MCWBR	6.9	9.8	12.0	13.7	13.7	13.9	13.5	13.6	12.3	10.4	8.1	6.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.291	0.315	0.348	0.381	0.392	0.398	0.399	0.394	0.369	0.341	0.313	0.287	(40)	
		taud	2.450	2.388	2.353	2.287	2.291	2.307	2.295	2.329	2.376	2.448	2.480	2.457	(41)	
	Ebn at Noon		790	843	866	868	870	865	857	846	832	801	754	747	(42)	
		Edh at Noon	64	85	103	122	126	125	125	116	100	80	62	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.04	1.86	3.03	4.28	4.98	5.66	5.39	4.72	3.47	2.16	1.16	0.84	(44)	
	RadStd		0.13	0.27	0.41	0.59	0.54	0.48	0.48	0.47	0.39	0.24	0.15	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+48	N/A
		N/A	N/A	N/A	N/A	+0.40	+0.36	N/A	N/A	+53	+17

Nomenclature: See separate page